UNIT-5 INDUSTRIAL PHARMACY-I

ONESHOT-CRASH COURSE

CARELESS PHARMA

IMPORTANT QUESTIONS

M.I. 1 — Define Aerosols and write its types. write a detailed note on :- (Aerosol System)

- Basic components of valve and aerosol containers + classify propellants. (then Importance + properties)
- M.M.I. — Quality control tests (evaluation parameter) for aerosols.

2 M.I. — Discuss about materials used in packaging of pharmaceutical products with special reference to Glass — Quality control test. / Adv. as a containers.

- factor influencing choice of pautical containers + sterility aspects.
- Legal and official requirements for packaging material.

3 Imp. — Define cosmetics. Write formulation and preparation of

- Lipsticks
- Vanishing cream/cold cream
- Shampoos.

Q. Define Aerosols and Write its types. Write a detailed note on Basic component of valve and aerosol containers, classify propellants.

- Quality control tests (Evaluation paramiter) for Aerosols.

AEROSOLS

Aerosols are pressurized pharmaceutical dosage forms that contain one or more active ingredients dispersed in a propellant - solvent mixture and are expelled as a fine mist, spray, aerosols or form through a metering or non - mrtering valve when actuated.

- They provide a non - touch, hygienic route for local or systemic delivery and are widely used in respiratory therapy, dermatology, nasal delivery and topical applications.

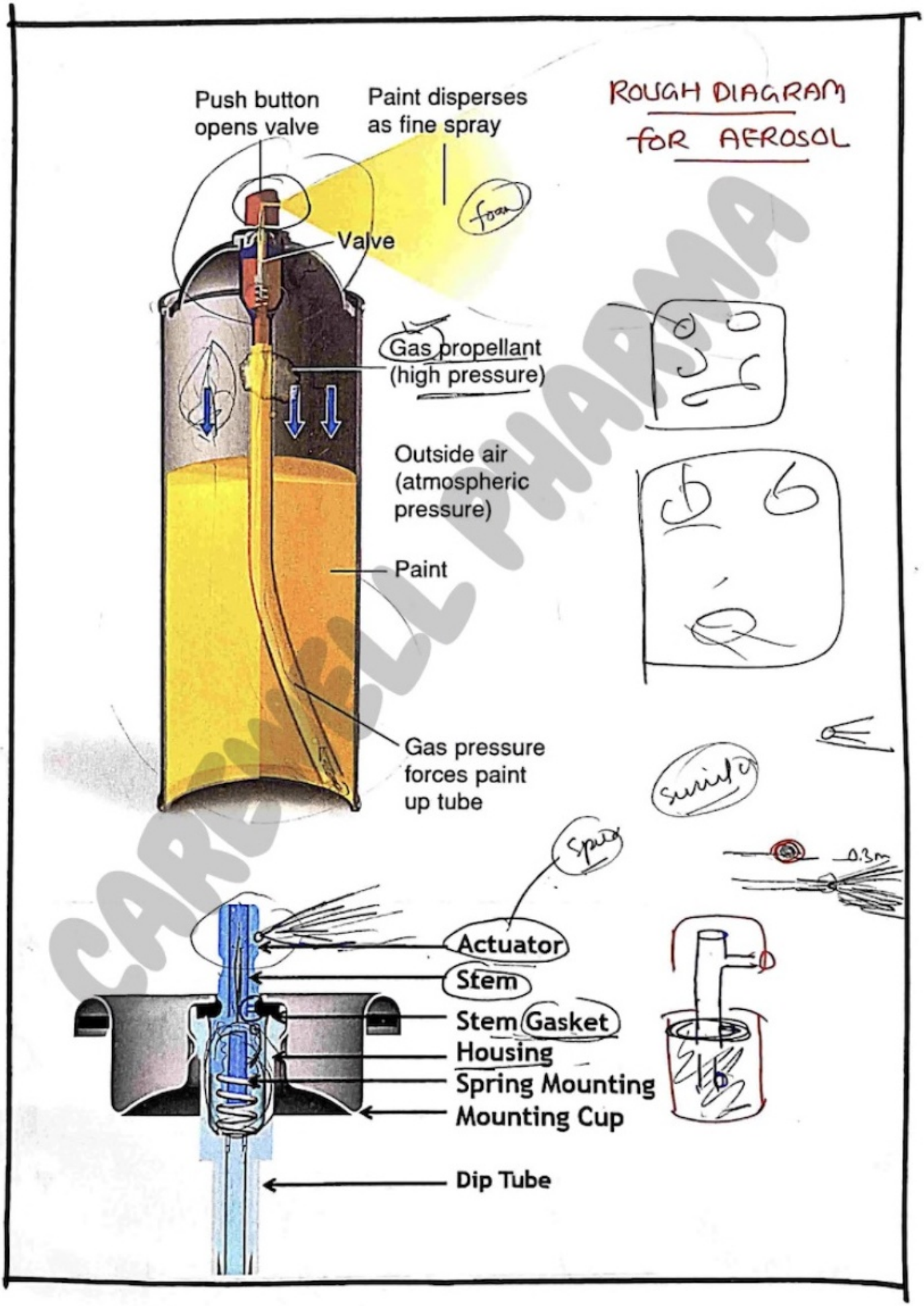
- They deliver medicine directly to the site of action (lungs, skin, nose) in a hygienic controlled way.

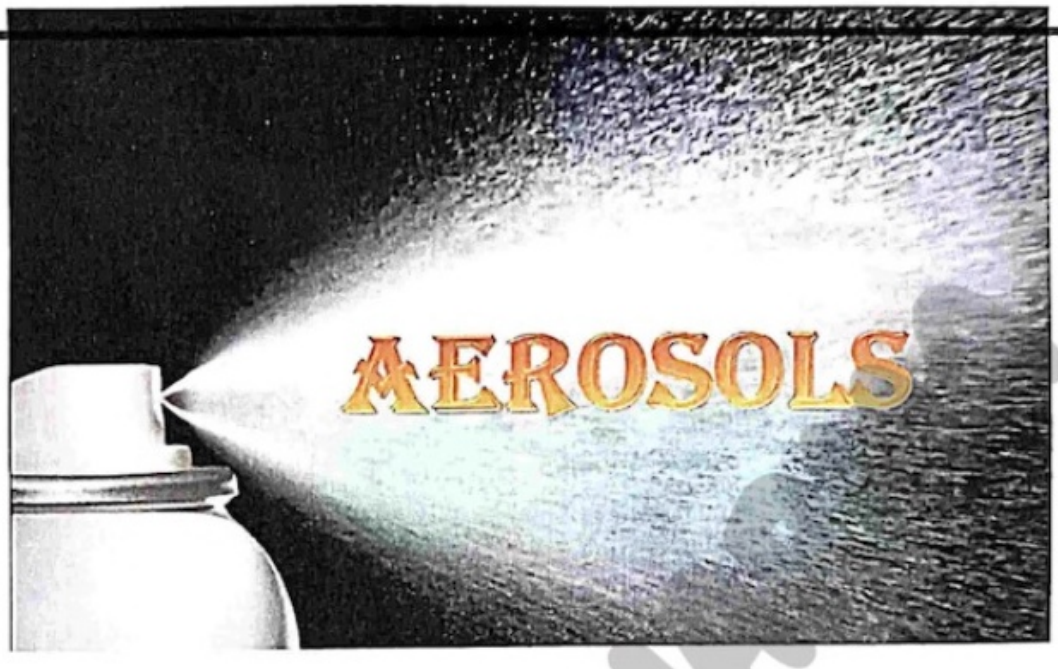
Eg. Inhales for asthma, nasal sprays etc....

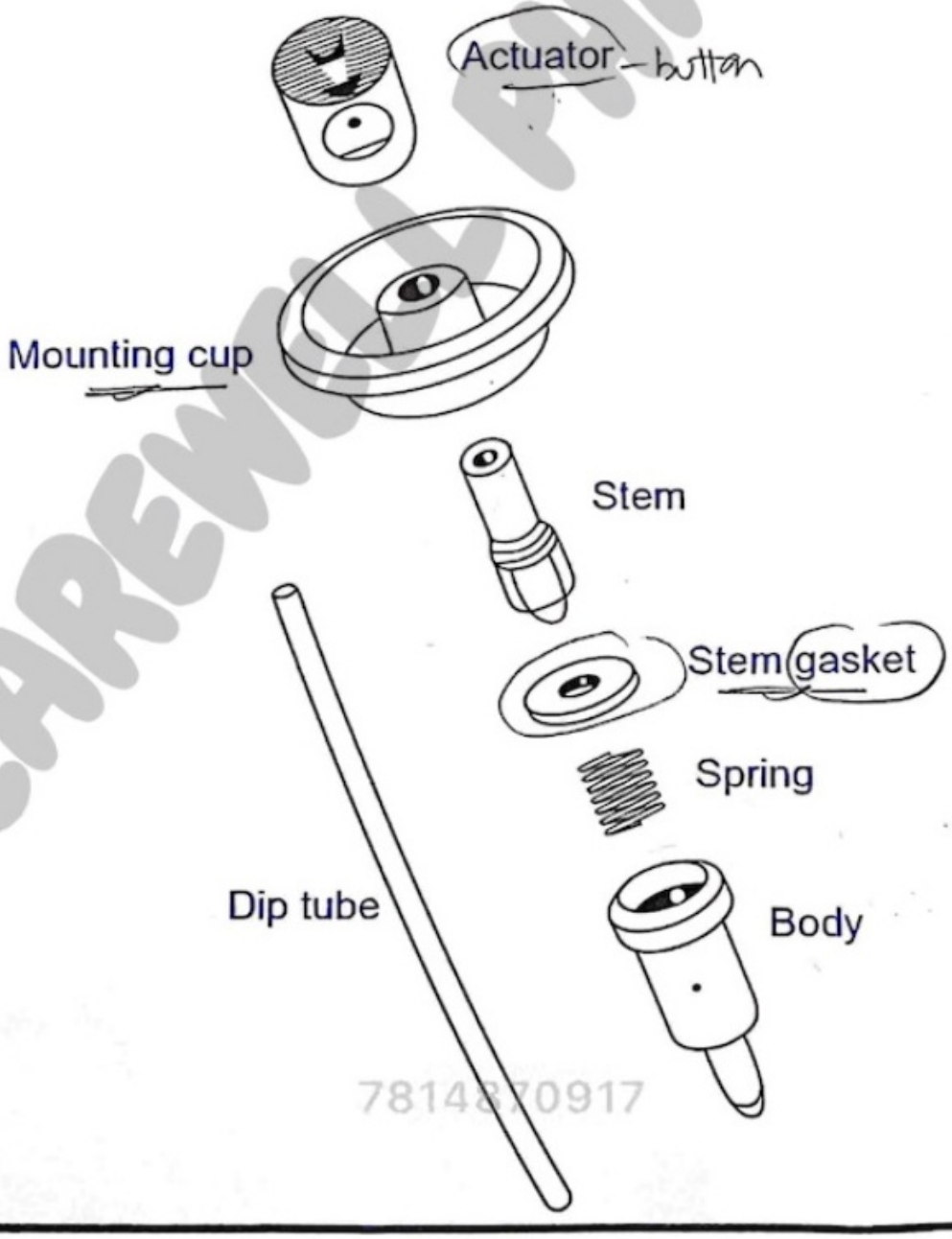
Objectives.

- Deposite drug directly at the site of action to increase local effect and reduce systemic exposure.
- Fine droplets / particle give quick absorption when needs.
- Non - contact administration reduces contamination and improves patient compliance.
- Metered dose designs provide uniform dose per actuation.

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ROUGH DIAGRAM FOR AEROSOL
Push button opens valve
Paint disperses as fine spray
Valve
Gas propellant (high pressure)
Outside air (atmospheric pressure)
Paint
Gas pressure forces paint up tube
Actuator
Stem
Stem Gasket
Housing
Spring Mounting
Mounting Cup
Dip Tube
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Advantages.

- Easy to use and portable.
- Rapid onset of action
- Protects drug from light, oxygen, moisture (sealed container).
- Accurate dose delivery (especially with metered values).
- Can deliver drug in differ forms: spray, foam, mist.
- Hygienic, no contamination (product not touched).

Disadvantages.

- High cost of packaging and valve system.
- Complex manufacturing process.

TYPES OF AEROSOL SYSTEM

Types	Explanation	Examples
(i) Two-Phase System	Contains drug concentrate + Liquefied propellant	Nasal sprays, Topical Antiseptics
(ii) Three-phase system	Contains aqueous phase + immiscible Liquids + propellants	some dermato-logical spray
(iii) Compressed gas system	Uses gases like CO_2, N_2 instead of liquefied propellant	foam cleaners, some medical spray
(iv) foam system	Emulsified product + propellants → forms foam on release	shaving cream, dermatological foams
(v) Metered Dose Inhaler (MDI)	special valve delivers fixed dose per (actuation.)	Asthma inhalers, COPD therapy.

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Basic Components of Aerosols System

An aerosol system is made up of 5 main components.

(i) Propellant
(ii) Valve
(iii) Containers
(iv) Actuator
(v) Product Concentrate.

(i) Propellant - These are gases or liquid that create pressure inside the aerosols container and help expel the drug as spray or foam.

- Also effects spray quality, particle size and drug delivery.
- Can be liquefied gases (Eg. butane, propane, HFAs) or compressed gases (Eg. CO_2, N_2).

Types of Propellants

A. Liquefied Gas Propellants - Store as liquid under pressure, vaporize on release.

- Provide constant pressure throughout use.

Eg. Hydrocarbons - Butane, propane, Isobutane.

B. Compressed Gas Propellants - Gases filled under high pressure, not liquefied.

- Pressure decreases as product is used.

Eg. Co_2, Nitrogen, Nitrous oxide.

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Functions of Propellant

- Create pressure to push drug out.
- Help in atomization (fine mist).
- Maintain spray quality and dose accuracy.

(ii) Valve - It is the control unit of the aerosol system.

- It is mechanical part that controls the release of drug from the aerosol container. It ensures safe, accurate and consistent dosing.
- It is the heart of the aerosols.

Parts of Valve System.

- Mounting cup - Attaches valve to container, seals it.
- Valve Housing - Holds metering chamber, controls flow.
- Valve Stem - Outlet path for drug; connects to actuators.
- Spring - Helps valve close after release.
- Dip Tube - Pulls drug from bottom of container.
- Metering Chamber - Stores fixed dose (used in MDI).
- Actuator - Button pressed by user to release spray.

Working of Valve - User presses actuator.

- Spring compresses, valve opens.
- Dip tube pulls drug up.
- Metering chamber release fixed dose.
- Drug exits via valve stem
- Spray forms, valve close automatically

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(iii) Container - The container is the body of the aerosol system It stores the drug and propellant under high pressure and must be strong, leak-proof and compatible with the formulation.

Materials Used.

- Aluminum - Lightweight, rust-proof, widely used.
- Tin-plated Steel - Strong, suitable for high-pressure.
- Glass - Chemically safe, used in low-pressure systems.

(iv) Actuator - The button or nozzle that user presses.

- It decides the spray pattern (fine mist, foam, stream).
- Connected to valve stem.
- Different shapes for different uses (nasal, oral, topical).

(v) Product Concentrate - The actual drug formulation inside the container.

- Can be a solution, suspension, emulsion or foam base.
- Includes active drug + excipients (solvent, stabilizer, surfactant).

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Quality control tests of aerosols

- These tests of aerosols are done to make sure that the aerosol works properly.
- These tests check that the spray comes out correctly, the pressure is safe, the dose is correct, and there is no leakage.
- It's main purpose is to check the safety, performance and consistency of the aerosol product before it reached the patient.

- It includes following tests :-
  - (i) Leak test / valve integrity test
  - (ii) Net content or weight variation test
  - (iii) Spray Pattern Test
  - (iv) ~~Particle or Droplet size test~~
  - (v) Dose per Actuation test
  - (vi) Pressure test

(i) Leak test :- This test is used to check that aerosol container is completely sealed or not. If there is any small leak, the aerosol will lose pressure.

Procedure → The can is placed in warm water. If bubbles come out, it means there is leakage.

- If the aerosol leaks, pressure become low and the spray will not work properly.

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(ii) Net content / weight variation test:- This test is used to check that how much product is present inside the can.

Procedure: weigh full and empty container, the difference in weight gives the actual amount of product inside.

- The can must have the exact amount written on the label.

(iii) Spray pattern test:- This test is used to check that how the spray looks when it comes out of the nozzle.

Procedure: Spray is released on a white paper or plate from a fixed distance. The round pattern formed is checked for size & uniformity.

- A proper spray pattern ensures that the drug spread evenly.

(iv) flame projection test:- This test is used to check that how flammable the aerosol is.

Procedure → Spray the aerosol toward a small flame and measure how much the flame extends or increases.

- It is important to know the fire risk, to ensure customer's safety, and to decide proper warning labels.

(v) Dose per Actuation test:- This test is used to checks how much drug comes out in one press (one actuation)

Procedure → Several sprays are collected in a solvent and tested by chemical analysis.

- It ensure that patient must receive the correct dose every time they use the spray.

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(VI) Pressure test: This test is used to measure the pressure inside the aerosol can.

Procedure → A pressure gauge is used to measure the pressure at room temperature.

- It is important because correct pressure is needed for proper spraying and for safety.

Evaluation Parameter

These are those test which are done during formulation development, to check where the aerosol formulation is working correctly or not before finalizing it.

- It includes :-
  - Physical Appearance
  - Density of formulation
  - Vapour pressure
  - foam stability
  - Particle / Droplet size
  - flammability check
  - sterility test etc.

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Manufacturing of Aerosols.

- It means filling the drug + propellant into a strong container with valve system so that it works safely under pressure. Since aerosols are pressurized, special filling techniques are used to avoid leakage and ensure dose accuracy.

(i) Cold Filling Method -

- Product concentrate (drug soln./suspension) cooled to -30 to -40°C.
- Liquefied propellent also cooled separately.
- Both are filled into the container at low temperature.
- Valve is sealed on the container.

Use - Suitable for non-aqueous systems (because water freezes).
Limitation - Not good for aqueous formulation; risk of moisture condensation.



(ii) Pressure Filling Method -

- Product concentrate filled into container at room temperature.
- Valve crimped (sealed).
- Propellent is added under high pressure through the valve.
- Container is tested for leaks and pressure.

Uses - Common method - suitable for aqueous and non-aqueous systems.
Advantage - Safer, Faster and Avoids Freezing Problems.

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Advantages -

- Ensures sterility (sealed system)
- Protects drug from light, oxygen, moisture.
- Provides accurate dosing with metered valves.
- Allows multiple forms (spray, mist, foam).
- Improves patient convenience and compliance.

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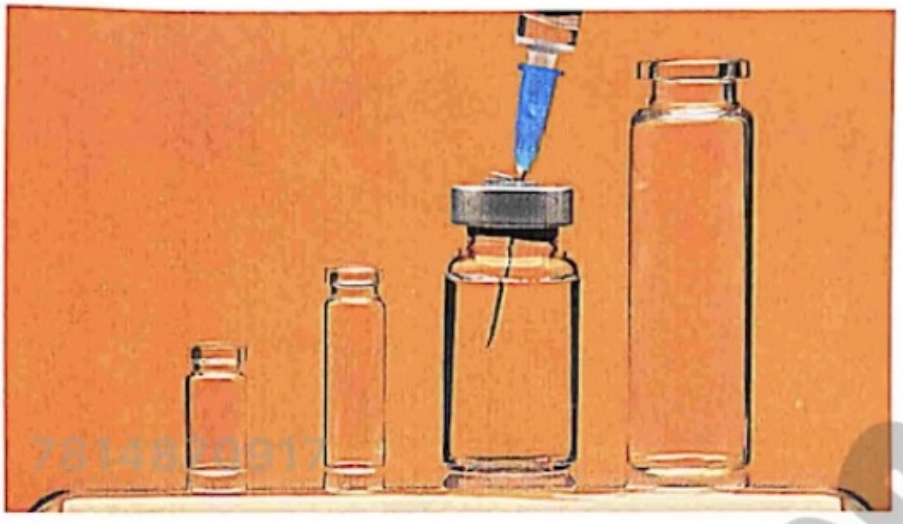
② Discuss about materials used in packaging of pharmaceutical product with special refrence to Glass and its quality control tests.

- Factor influencing choice of pharmaceutical containers.
- Legal and official requirements for packaging material.

Pharmaceutical Packaging - Packaging in pharmacy means enclosing drugs in suitable containers to protect, preserve and deliver them safely.

- Its main purpose is to protects medicine from light, moisture, oxygen, contamination and mechanical damage.

eg. • Tablets in blister packs.
  - Syrup in glass or plastic bottles
  - Injections in glass vials or ampoules.
  - Ointements in aluminum tubes.

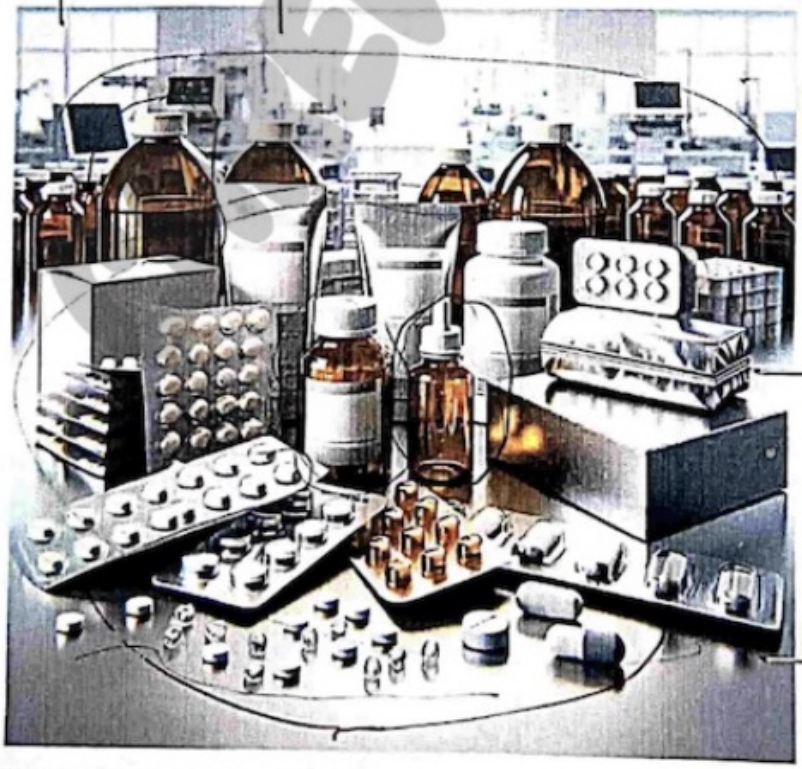


ROUGH IMAGES FOR PACKAGING MATERIAL (GLASS)



Ampals



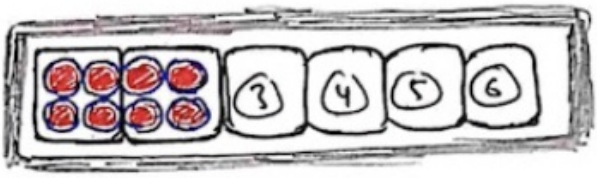




Vials

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Types of Packaging.



(i) Primary packaging - Direct contact with drug (bottle, blister, vial).
(ii) Secondary packaging - Outer cover for grouping (cartons, boxes).
(iii) Tertiary packaging - Bulk handling and transport (shipping containers).

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Materials used in Pharmaceutical Packaging.

(i) Glass - inert, transparent, used for injectables and liquids
(ii) Plastics - Lightweight, Flexible, used for bottles, IV bags, blister packs.
(iii) Metals - Aluminum, tin, used for tubes, foils, aerosols cans.
(iv) Rubber - Closures and stoppers for vials, provides airtight seal.

Glass as a Packaging Material

- It is considered the gold standard for packaging because it is non-reactive, transparent and provides excellent protection against environmental factors. It is mainly used for ampoules, vials, bottles and syringes.

Composition of Glass

- Made from silica (SiO_2) as the main ingredients.
- Mixed with soda (Na_2O), lime (CaO), alumin (Al_2O_3) and boric oxide (B_2O_3).
- The exact composition decides the type of glass and its resistance properties.

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Types of Glass

① Type I - Borosilicate Glass.-

Composition - High percentage of silica (SiO_2) with boric oxide (B_2O_3), alumina, and alkali oxides.

Properties -

- Very high hydrolytic resistance. (does not leach alkali)
- Resistant to thermal shock and sudden temperature changes.
- Chemically inert, suitable for sensitive drugs.

Uses - Solutions that are highly reactive or unstable.

- Parenteral (injectable) products like vaccines, insulin, antibiotics.

Example - Clear or amber borosilicate vials and ampoules.

② Type - II Treated Soda - lime Glass.

Composition - Ordinary soda - lime glass treated internally with sulfur or other agents to reduce alkali leaching.

Properties -

- Moderate hydrolytic resistance (better than Type III).
- Surface treatment improves chemical stability.
- Cheaper than borosilicate but still safe for many injectables.

Uses -

- Acidic and neutral aqueous solutions for parenteral use.
- Injectable drugs that are not highly alkaline.

Example - Vials for injections like saline, glucose solutions.

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3. Type III - Regular Soda - lime Glass.

Composition - Soda - lime glass without special treatment.

Properties -

- Lower hydrolytic resistance compared to Type I and II.
- Can leach (alkali) into solution if used for injectables.
- Suitable for solid dosage forms.

Uses -

- Oral solid dosage forms like tablets and capsules.
- Dry powders that are not sensitive to moisture.

Examples - Glass bottles for (tablets), capsules, syrups.

4. Type NP - Non - Parenteral Glass.

Composition - Similar to Type (III) soda - lime glass, but specifically designed for non - injectable products.

Properties -

- Lowest hydrolytic resistance.
- Not suitable for parenterals.

Uses -

- Topical and oral products like ointments, lotions, eye drops (non - injectable).
- Cosmetic and general pharmaceutical liquids.

Example - Glass (jars) for ointments, eye drop bottles.

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Advantages

- It is chemically inert - (no reaction with drug)
- It is Transparent - easy Inspection of contents.
- It is Impermeable - protects from moisture, gases and vapors.
- It is sterilized - can withstand autoclaving and dry heat.
- Recyclable - eco-friendly.

Disadvantage

- It can (break) easily.
- More expensive than plastic packaging.
- Difficult to transport compared to plastics.

Stability Aspects of Glass Containers

(i) Chemically Inertness - Glass (especially type 1) does not react with drug → maintain purity.

(ii) Moisture & Glass Barrier - Glass is non-porous → protects from moisture and oxygen.

(iii) Light Protection - Amber glass blocks (UV) light → prevents degradation of light-sensitive drugs.

(iv) Thermal Stability - Borosilicate glass tolerates autoclaving and temperature changes.

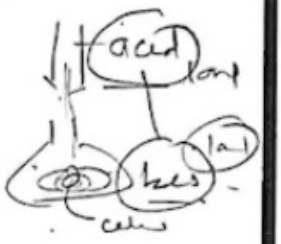
(v) Leaching Stability - Type 1 glass has low alkali leaching → ideal for injectables.

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Quality Control Tests for Glass Containers

(i) **Hydrolytic Resistance Test** - Its main purpose is to measure how much alkali (Na^+, K^+) leaches from glass into water.

- Leached alkali can degrade the drug or change pH.



Method -

1. Fill glass container with purified water.
2. Autoclave at 121°C for 30 minutes.
3. Cool and titrate with acid using methyl red indicator.

Result -

- Type I - very low alkali → passes easily.
- Type III - Higher alkali → not suitable for injectables.

(ii) **Thermal Shock Test** - Its main purpose is to checks resistance to sudden temperature changes.

- Prevents cracking during sterilization or transport.

Methods -

1. Heat glass container to 100°C
2. Immediately immerse in cold water (~ 20°C)
3. Observe for cracks or breakage.

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Result - Type I glass should remain intact.

(iii) Fragmentation Test :- Its main purpose is to ensures glass breaks safely if damaged.

- Prevents injury from sharp fragments.

Method

1. Drop test or mechanical impact
2. Observe size and shape of broken pieces.

Results - It should break into dull - edged, fewer fragments.

(iv) Transparency and Clarity Test. Its main purpose is to confirms visual inspection of contents is possible.

- Detects particles, discoloration or contamination.

Methods

1. Inspect under bright light
2. Checks for bubbles, cracks, haze.

Results - It should be clear and defect - free.

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(v) Surface Chemical Resistance Test (for type II) - Its main purpose is to checks durability of surface-treated soda-lime glass.

- Ensures surface treatment remains effective.

Method

1. Apply acid or alkali to surface
2. Analyze for reaction or degradation.

Result - It should show minimal surface change.

Plastic as a Packaging Material

- Plastics are synthetic polymers used extensively in packaging due to their low cost, versatility and resistance to breakage. They can be molded into bottles, blister packs, IV bags, closures and films.
- They must be carefully selected because some plastics can interact with drugs (adsorption, leaching, permeability).

Types of Plastics

1. Polyethylene [PE] - It is flexible, moisture-resistant, chemically inert.

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- It is two types: Low density PE (LDPE), High density PE (HDPE).
- LDPE - squeeze bottles, dropper bottles.
- HDPE - rigid bottles for tablets, capsules, syrups.

Limitation - Poor barrier to oxygen and volatile oils.

2. **Polypropylene (PP)** - It is high heat resistance, good chemical resistance, lightweight.
- It is used in containers for autoclaved products.
- It also used as closures, IV bottles, syringes.

Limitation - Brittle at low temperature.

3. **Polyvinyl Chloride (PVC)** - It is Transparent, strong, good barrier to oils.
- It is used in blister packs for tablets and capsules.
- It is used in IV bags, flexible tubing.

Limitations - Can release plasticizers (like phthalates), not suitable for sensitive drugs.

4. **Polystyrene (PS)** - It is clear, rigid, lightweight.
It is used as Disposable containers, diagnostic kits.

Limitations - Poor resistance to solvents, brittle.

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5. **Polyethylene Terephthalate (PET)** - It is Transparent, strong, good barrier to gases.

- It is used as bottles for liquids (syrups, oral solutions).

Limitations - More expensive than PE/PP.

Advantages of Plastics.

- Lightweight and unbreakable compared to glass.
- Easy to mold into different shapes.
- Patient-friendly (easy handling).
- Resistant to moisture.

Metals as a Packaging Materials

- It is strong, opaque, excellent barrier to light, moisture and gases.
- It is used as collapsible tubes (ointments, creams), foil strip for tablets, aerosols cans.



Example - Aluminum, tin, stainless steel.

Limitation - Can react with some drugs, so protective coatings are often required.

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Rubber

- It is flexible, provides airtight and leak-proof seal.
- Used as closures for vials, stoppers for injectables, gaskets in caps.

Example - Natural rubber, Silicon rubber, butyl rubber.

Limitation - May absorb or adsorb drug components, requires compatibility testing.

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Fators Influencing Choice of Pharmaceutical Containers

- Pharmaceutical containers are selected to ensure drug stability, safety, and patient convenience. Since different dosage forms and drugs have different sensitivities, the container must be chosen carefully to avoid degradation, contamination or dosing errors.

(1) Nature of Dosage Form.

- Liquids need leak-proof, inert containers.
- Solid need moisture-resistant packaging.

Example - Syrup → plastic / glass bottles,
Tablets → Blister packs.

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② Physiochemical Properties of Drug.

- Sensitivity to light, moisture, oxygen, pH or temperature.

Example - Light-sensitive drugs → amber glass bottles,
Moisture-sensitive tablets → foil blister packs.

③ Route of Administration.

- Parenterals need sterile, pyrogen-free containers.
- Orals need basic protection

Example - Injections → Type I glass vials,
Capsules → HDPE bottles.

④ Compatibility with Container Material.

- Container must not react with or absorb drug components.

Example - PVC may leach plasticizers → not suitable for sensitive liquids.

⑤ Barrier Properties -

- Must protect against moisture, gases and light.

Example - Aluminum foil provides excellent moisture and light barrier.

⑥ Sterilization Requirements.

- Container should tolerated autoclaving, dry heat or radiation.

Example - PP bottles withstand autoclaving; LDPE does not.

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7. Patient Convenience and Handling.

- Easy to opin, carry and use.

Example - Child - resistant caps for pediatric syrup.

8. Cost and Availability.

- Material should be economical and easily available.

Example - Plastics are cheaper and widely used for oral products.

9. Regulatory and Pharmacopoeial Standards.

- Must comply with IP, USP, BP, well-closed, air-tight, light-resistant, etc..

Example - Injectables must pass sterility and hydrolytic resistance tests.

- The choice of container depends on dosage form, drug properties, route, compatibility, barrier needs, sterilization, patient ease, cost and legal standards. A well-chosen container ensures drug safety, stability and compliance.

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Legal and official Requirements for Packaging Materials

- Packaging materials are regulated to ensure they protect the drug, are safe for patients, and comply with legal standards. Since packaging comes in direct contact with the drug, it is considered part of the dosage form and must meet strict official requirements.

1. Protection of Product.
   - Must protect drug from light, moisture, oxygen and contamination.

   Example - Amber glass for light-sensitive drugs; foils packs for moisture sensitive tablets.

2. Inertness and Non-toxicity.
   - Material must not react with drug or leach harmful substance

   Example - Type I glass is inert and safe for injectable.

3. Compliance with Pharmacopoeial Standards.

   IP, USP, BP define container types:
   - Well-closed → protects from contamination.
   - Air-tight → prevents entry of air / moisture.
   - Light-resistant → blocks harmful light.
   - Single-dose / Multi-dose → defined for injectables.

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④ Quality Control Tests

- Containers must pass official tests:
- Sterility (Injectables)
- Leachables and extractables (rubber, plastics).

⑤ Labelling and Identification.

- Must carry proper label: drug name, strength, batch no., expiry, storage.

Example - Syrup bottles labeled with "Shake well before use"

⑥ Sterility and Safety (for Parenterals)

- Must be sterile, pyrogen-free, tamper-proof.

Example - Rubber stoppers and glass vials used for vaccines.

⑦ Environmental and Recycling Norms.

- Should comply with eco-friendly standards.

Example - Avoid banned CFC propellants; use recyclable glass / plastics.

⑧ Regulatory Approval.

- Materials must be approved by authorities like FDA, CDSCO, WHO.

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③ Define cosmetics. Write formulation and preparation of
- Lipsticks, Vanishing cream, Cold cream, shampoos, toothpaste, Hair dyes, Sunscrens.

Cosmetics

- These are external preparation applied to the human body (skin, lips, hair, nails) mainly for cleansing, beautifying, promoting attractiveness or altering appearance.
- They are not medicines but play an important role in personal care, grooming and confidence building.

eg. Vanishing cream, Lipsticks etc...

Advantages.

- Improve appearance and self-confidence.
- Provide cleansing and protective effects.
- Easy to apply, widely available, and patient-friendly.
- Some cosmetics also have moisturizing or conditioning properties.

Disadvantages

- Expensive branded products may not be affordable.
- May cause allergic reactions or irritation in sensitive individuals.

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Lipsticks They are solid sticks applied on lips to give color, shine and protection. They are one of the most popular makeup products used daily. A good lipstick should be smooth, glassy, stable and safe for lips.

Formulation -

- Waxes (Beeswax, Carnauba, Candelilla) - hardness and structure.
- Oils (Castor oil, Mineral oil, Lanolin) - smoothness and shine.
- Pigments (Iron oxides, Titanium dioxide, organic dyes) - color.
- Preservatives (BHA, BHT) - prevent microbial growth.
- Perfume / Flavor - improve acceptability.

Preparation.

- Melt waxes and oils together at 70-80°C
- Add pigments slowly and mix until uniform dispersion.
- Add perfume when mixture cools slightly.
- Pour into lipstick molds, cool and solidify.
- Remove, flame surface for shine and pack in lipstick cases.

- Lipsticks should have proper melting point (55-65°C), hardness, smooth surface, and stability (no sweating or color change).

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O/W emulsion

Vanishing Cream These are oil-in-water emulsions that disappear quickly after application, leaving a thin, invisible film. They are non-greasy, light and mainly used as foundation base or for oily skin.

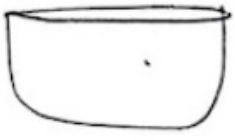


Formulation. → oil → oil phase

- Stearic acid - Base, gives body.
- Alkali (NaOH / KOH) - emulsifier with stearic acid.
- Glycerin - humectant, prevents dryness. → moisturiser
- Water - Continous phase → aqueous phase
- Perfume / Preservatives - improves acceptability.

Preparations.

- Melt stearic acid and oily ingredients.
- Dissolve alkali and glycerin in water (aqueous phase).
- Add aqueous phase to oily phase with stirring → O/W emulsion
- Cool mixture, add perfume, homogenize
- Pack in jars or tubes.



- Called "Vanishing" because cream disappears after rubbing.
- Non-greasy, suitable for oily skin, used as base for makup.

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w/o emulsion

Cold Cream. These are water-in-oil emulsions that feel greasy and provide moisturizing and cleansing effect. They are especially useful for dry skin and give a cooling sensation when applied.



w/o em

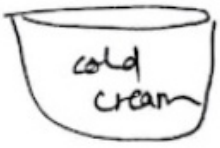
Formulation.

- Beeswax - Base, gives consistency.
- Mineral oil / Light liquid paraffin - oil phase, emollient.
- Borax - emulsifier (react with fatty acids)
- Water - internal phase
- Perfume / Preservatives - improve acceptability.

Preparation -



- Melt beeswax and mineral oil (oil phase) :-
- Dissolve borax in water (aqueous phase) :-
- Add hot aqueous phase to oil phase with stirring - (w/o) emulsion
- Cool mixture, add Perfume.
- Pack in wide-mouth jars.



- Cold creams are greasy, used for dry skin, cleansing and for dry skin, cleansing and moisturizing. Provide cooling effect when applied.

CADEWELL PHARMA

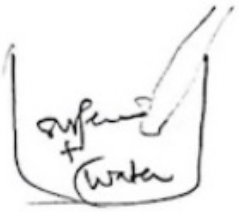
Shampoos - These are (liquid) cosmetic preparations used for cleaning hair and scalp. They contain (surfactants) that remove dirt and oil, along with conditioners to improve (hair) texture.

- A good shampoo should foam well, clean effectively, and be mild to scalp.

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Formulation.

- Surfactants (Sodium lauryl sulfate, ammonium lauryl sulfate) - (cleansing.)
- Conditioners (Silicones, proteins) - improves hair texture.
- Thickeners (CMC, PEG) - adjust viscosity.
- Preservatives (Parabens) - prevents microbial growth
- Perfume / Color - improve acceptability.
- Water - vehicle.



Preparation.

- Dissolve (surfactants) in water to form base solution
- Add conditioners and thickeners with stirring.
- Mix preservatives and perfume.
- Adjust pH to (5-7) (mild for (scalp))
- Fill into bottles with dispensing caps.



- Shampoos should produce good foam, clean hair effectively and be non-irritant types include medicated shampoo, baby shampoos.

CAREWELL PHARMA

Toothpaste These are semi-solid preparation used to clean teeth, remove plaque and freshen breath. They may be therapeutic (fluoride-based) or cosmetic (whitening, herbal).



Formulation.

- Abrasives (Calcium carbonate, silica) - Remove debris and stains.
- Humectants (Glycerin, sorbitol) - Prevent drying.
- Surfactants (Sodium lauryl sulfate) - Foaming and cleansing.
- Binders (CMC, xanthan gum) - Maintain consistency.
- Fluoride compounds (NaF, MFP) - Prevents cavities.
- Flavoring agents (Mint oil) - Improve taste.
- Preservatives and sweeteners - Stability and palatability.

Preparation.



- Mix humectants and binders to form base
- Add abrasive and surfactants.
- Incorporate fluoride, flavor and preservatives.
- Homogenize and deaerate.
- Fill into laminated tubes.

- Should be smooth, non-irritant, pH 6-8, free from grittiness.
- Therapeutic toothpaste must meet fluoride content standards.

CARMEL PHARMA

Hair Dyes These are cosmetic products used to colour hair temporarily or permanently. They may be chemical (oxidative) or herbal (henna). Permanently dyes involve oxidation reactions.

Formulation.

- Color precursor (Paraphenylenediamine - PPD) - React to form color.
- Couplers (Resorcinol, naphthol) - Modify shade.
- Oxidizing agents (Hydrogen peroxide) - Develop color.
- Conditioners (Silicones, proteins) - Protect hair.
- Solvents (Water, ethanol) - Vehicle.
- Preservatives and fragrance - Stability and acceptability.



Preparation.

Prepare dye base with precursor and couplers → mix with oxidizer before use → apply to hair → leave for 20-30 minutes → rinse and condition.

- Patch test required before use. Should not damage hair or scalp
- Permanent dyes last 4-6 weeks, semi-permanent dyes fade in 8-10 washes.

CARSWELL PHARMA
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Sunscreens These are topical preparations that protect skin from harmful UV rays, preventing sunburn, tanning and premature aging. They are classified by SPF (Sun Protection Factor)

Formulation.

- UV Filters (Zinc oxide, titanium dioxide, oxybenzone) - Block or absorb UV rays.
- Base (Creams, lotion, gel) - Vehicle for application.
- Moisturizers (Aloe vera, glycerin) - Hydration.
- Antioxidants (Vitamin E, C) - Skin protection.
- Preservatives and fragrance - Stability and appeal.

Preparation -

Disperse UV filters in base → add moisturizes and antioxidants → mix preservatives and perfume → homogenize and adjust pH → fill into tubes or bottles.

- SPF 15-50 commonly used. Should be water-resistant, non-comedogenic and dermatologically tested. Apply 15-30 minutes before sun exposure.

PARSHELL PHARMA
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